Ansible Overview:

Ansible is an open-source automation tool used to manage IT infrastructure by automating tasks like configuration management, application deployment, and orchestration.

What Ansible Does?

Automates repetitive tasks

From installing packages to configuring services, Ansible handles it all.

Manages system configurations

Ensures servers and environments stay consistent across deployments.

Deploys applications

Supports rolling updates and zero-downtime deployments.

Orchestrates complex workflows

Coordinates multi-tier applications and services across environments.

Key Features

Agentless architecture:

No need to install software on managed nodes; it uses SSH or WinRM.

Human-readable playbooks:

Written in YAML, playbooks describe desired system states in a clear, declarative format.

Idempotent execution:

Running the same playbook multiple times won't change the system if it's already in the desired state.

Extensible modules:

Thousands of built-in modules for tasks like file manipulation, service control, cloud provisioning, and more.

Real-World Use Cases

- Provisioning servers with specific OS and packages

- Enforcing security policies across fleets

- Deploying containerized apps with Kubernetes

- Automating CI/CD pipelines

- Managing network devices and firewalls

Ansible Architecture:

- **There are two types of machines in ansible architecture:**
- **1: Control Nodes:**
    - Ansible is installed and run from control nodes.
    - Control nodes should be Linux or Unix systems.
    - Microsoft windows are supported as Managed hosts.
    - It also has copies of your ansible project files.
    - A control node could be an administrator's laptop, a system shared by number of administrators or a server running RedHat Ansible Tower.
- **2: Managed Hosts:**
    - Are nodes managed by control nodes.
    - These could be windows, Linux or cloud-based nodes or network devices such as router & switches.

MANAGED HOSTS

- One of the benefits of Ansible is that managed hosts do not need to have a special agent installed.
- The Ansible control node connects to managed hosts using a standard network protocol to ensure that the systems are in the specified state.
- Managed hosts might have some requirements depending on how the control node connects to them and what modules it will run on them.
- Linux and UNIX managed hosts need to have Python 2 (version 2.6 or later) or Python 3 (version 3.5 or later) installed for most modules to work.

- For Red Hat Enterprise Linux 8, you may be able to depend on the platform-python package. You can also enable and install the python36 application stream (or the python27 application stream).

  [root@host ~]# yum module install python36

- If **SELinux** is enabled on the managed hosts, you also need to make sure the python3-libselinux package is installed before using modules that are related to any copy, file, or template functions. (Note that if the other Python components are installed, you can use Ansible modules such as yum or package to ensure that this package is also installed.)

Microsoft Windows-based Managed Hosts:

- Hosts Ansible includes a number of modules that are specifically designed for Microsoft Windows systems. These are listed in the Windows Modules [https://docs.ansible.com/ansible/latest/ modules/list_of_windows_modules.html] section of the Ansible module index.
- Most of the modules specifically designed for Microsoft Windows managed hosts require PowerShell 3.0 or later on the managed host rather than Python. In addition, the managed hosts need to have PowerShell remoting configured. Ansible also requires at least .NET Framework 4.0 or later to be installed on Windows managed hosts.

Managed Network Devices :

- You can also use Ansible automation to configure managed network devices such as routers and switches. Ansible includes a large number of modules specifically designed for this purpose. This includes support for Cisco IOS, IOS XR, and NX-OS; Juniper Junos; Arista EOS; and VyOS-based networking devices, among others.
- Because most network devices cannot run Python, Ansible runs network modules on the control node, not on the managed hosts. Special connection methods are also used to communicate with network devices, typically using either CLI over SSH, XML over SSH, or API over HTTP(S).

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